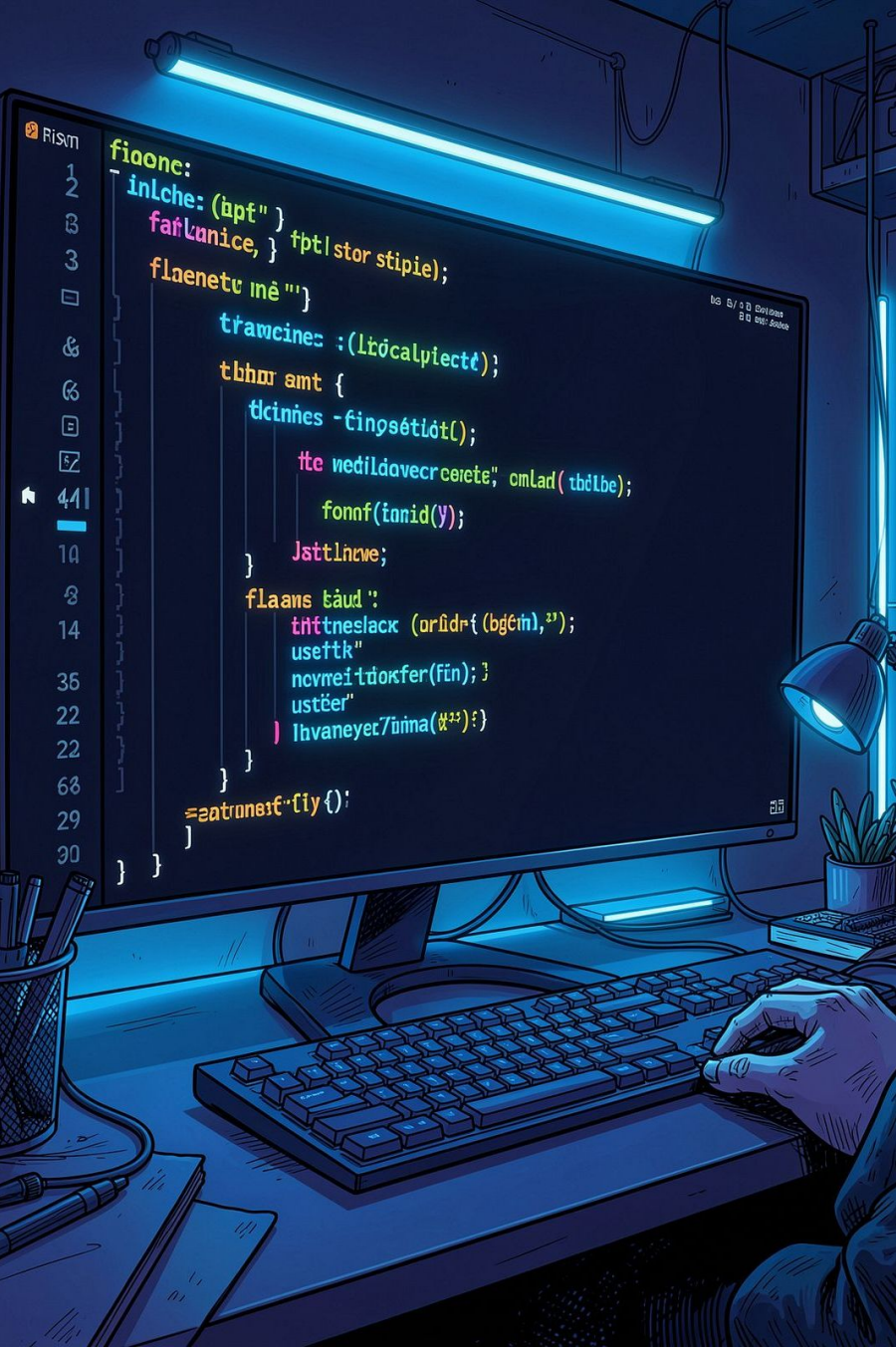




C++ Programming Virtual Internship Report

Submitted to School of Computer Science and Engineering,
IILM University, Greater Noida in partial fulfilment of B.Tech
CSE

RAJKUMAR

ROLL NO: 25SCS1003000803

BATCH 2025-29

Declaration & Acknowledgement

Candidate's Declaration

Rajkumar declares this report is original, not submitted elsewhere, and all references are properly cited. Solely responsible for any copyright violations.

Acknowledgement

Gratitude to Thiranex (Skill Development & Future Tech) for the structured platform, IILM University faculty for continuous support, and parents for their blessings.



Internship Certificates



Certificate of Achievement

Issued by Thiranex to Rajkumar for successful completion of C++ Programming Internship from 02 Aug 2026 to 01 Sept 2026.

Verified Certificate ID: THX-AUG0226-220

Signed by Hariharan M, Founder & CEO

Internship Offer Letter

Remote / Project-Based internship offered by Thiranex dated 02 August 2026. Intern ID: THX-AUG0226-220. Role: Intern – C++ Programming.



An illustration of a desk setup. A laptop is open, displaying a code editor with a dark theme and syntax-highlighted code. A desk lamp is positioned to the left, casting light on the desk. A window in the background shows a night scene with a plant on the sill. A notebook and pen are on the desk next to the laptop.

Introduction & Organisation Profile

Internship Overview

Conducted remotely at **Thiranex** from 02 Aug – 01 Sept 2026. Combined self-paced learning, mentor reviews, and a console-based mini project.

About Thiranex

A skill-development & future-tech organisation (MSME registered) offering remote, project-based internships with verifiable digital certificates. Led by **Hariharan M**, Founder & CEO.

Problem Statement & Objectives

The Gap

C++ is taught theoretically in most CS curricula, with limited opportunity to apply OOP, dynamic memory, and STL in a complete working application.

Key Objectives

- Master C++ fundamentals: syntax, control structures, functions
- Implement OOP: classes, inheritance, polymorphism
- Understand pointers and dynamic memory management
- Gain hands-on STL experience (vectors, maps, iterators)
- Build a console-based mini project with file handling

Scope & Technologies Used

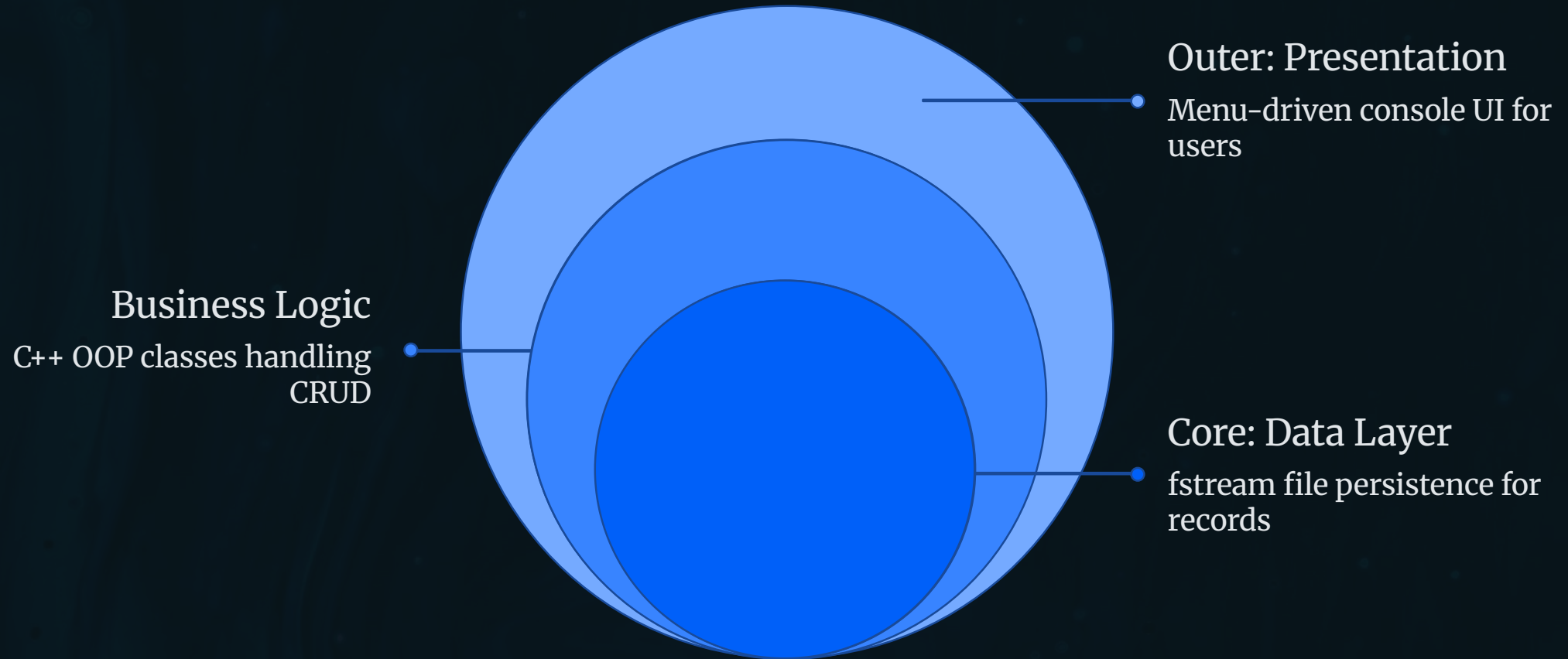
Scope of Project

Covers core-to-intermediate C++: fundamentals → OOP → pointers/memory → STL → file handling. Consolidated via a **Student Record Management System** (add, view, search, update, store records). Limited to virtual, self-paced environment — no live deployment.

Technologies & Tools

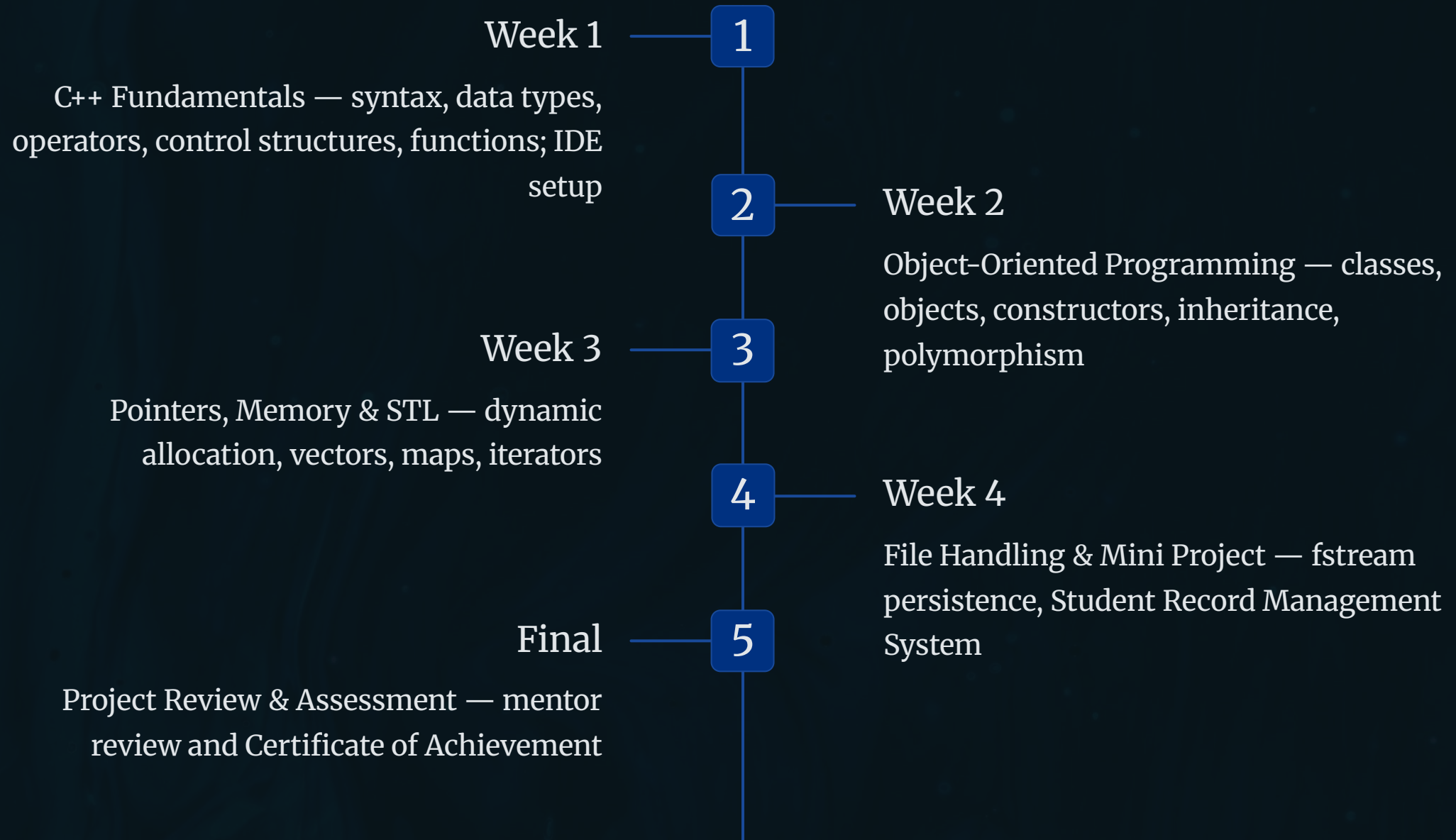
- **Language:** C++ (C++11/14)
- **Compiler:** GNU G++ (GCC)
- **IDE:** VS Code / Code::Blocks
- **Libraries:** STL – vector, map, string, algorithm
- **File Handling:** fstream
- **Version Control:** Git

System Architecture



The console-based Student Record Management System follows a clean layered design — separating user interaction, business logic, data persistence, and utility functions for maintainable, modular code.

Methodology — Weekly Breakdown



Expected Outcomes



Core C++ Proficiency

Write, compile, and debug C++ programs using core language constructs and OOP principles.



File Handling & Project

Hands-on file persistence experience; completed Student Record Management System demonstrating all skills.



Memory & STL Mastery

Working knowledge of pointers, dynamic memory management, and STL containers for efficient code.



Certified Completion

Verifiable Certificate of Achievement from Thiranex (ID: THX-AUG0226-220).

Bibliography & References

Primary Sources

1. Thiranex, "Offer Letter for Internship in C++ Programming," Intern ID: THX-AUG0226-220, 2026
2. Thiranex, "Certificate of Achievement – C++ Programming Internship," Verified ID: THX-AUG0226-220, 2026
3. B. Stroustrup, *The C++ Programming Language*, 4th ed., Addison-Wesley, 2013

Online References

1. cplusplus.com Reference — cplusplus.com/reference/
2. cppreference.com — en.cppreference.com
3. GeeksforGeeks, "C++ Programming Language" — [geeksforgeeks.org/c-plus-plus/](https://www.geeksforgeeks.org/c-plus-plus/)
4. Thiranex — thiranex.in

☐ All references have been quoted and cited. This report is original and has not been submitted to any other institute.